

(MATH200) Differential Equations

Problems

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Problems are arranged chronologically by year and semester, with the midterm preceding the final examination within the same semester, while preserving the original problem order within each source.

Problem 1. (2021 Spring KAIST Differential Equations and Applications) Answer the following questions.

- (a) Solve the initial-value problem

$$\frac{dy}{dx} = \frac{(50 - y)(y - 10)}{40}, \quad y(0) = 18.$$

- (b) Sketch the solution curve.
 (c) Find the point on the solution curve where dy/dx is maximized.

Problem 2. (2021 Spring KAIST Differential Equations and Applications) Solve the differential equation

$$(x + 1)y' + (x + 2)y = 2xe^{-x}, \quad x > -1.$$

Problem 3. (2021 Spring KAIST Differential Equations and Applications) Solve the initial-value problem

$$y^2 \cos(xy) \, dx - (2 - 2 \sin(xy) - xy \cos(xy)) \, dy = 0, \quad y(\pi) = 2.$$

Problem 4. (2021 Spring KAIST Differential Equations and Applications) Find the general solution of the differential equation

$$y'' - \frac{1}{x}y' = -x \sin x, \quad x > 0.$$

Problem 5. (2021 Spring KAIST Differential Equations and Applications) Consider the population model for a species of fish in a lake

$$\frac{dP}{dt} = 2P - \frac{P^2}{50},$$

where P is measured in thousands of fish and t is measured in years. The managing office of the lake wants to issue fishing licenses for the harvesting of some of the fish; this amounts to a constant term being subtracted off of the right-hand side of the above equation, which is a function of h , the number of licenses issued. Each fishing license is valid for the annual take of 3,000 fish. Answer the following questions.

- (a) What is the largest number of licenses that can be issued if the goal is to keep a stable population of fish in the lake over the long term? The number of licenses is, of course, a nonnegative integer.
 (b) If the largest number of licenses is actually issued, what is the expected long-term stable population of fish in the lake?

Problem 6. (2021 Spring KAIST Differential Equations and Applications) Consider the linear differential equation

$$ay'' + by' + cy = g(t),$$

where a, b, c are positive constants and $g(t)$ is a continuous function. If $y_1(t)$ and $y_2(t)$ are both solutions of the above equation, show that

$$\lim_{t \rightarrow \infty} (y_1(t) - y_2(t)) = 0.$$

Is this still true if $b = 0$?

Problem 7. (2021 Spring KAIST Differential Equations and Applications) Verify that $y_1(t) = 1$ and $y_2(t) = \sqrt{t}$ both solve the differential equation

$$yy'' + (y')^2 = 0, \quad t > 0,$$

but $y(t) = c_1 + c_2\sqrt{t}$ is not a general solution of the differential equation. Explain why this result does not contradict the superposition principle for homogeneous equations we have learned in the class.

Problem 8. (2021 Spring KAIST Differential Equations and Applications) Find an integer n such that $\phi(x) = \sin(x^n)$ satisfies the differential equation

$$xy'' - y' + 4x^3y = 0, \quad x > 0.$$

Find the general solution of the equation.

Problem 9. (2021 Spring KAIST Differential Equations and Applications) Find a particular solution of the differential equation

$$y'' - 2y' + y = e^x \arctan x.$$

Problem 10. (2021 Spring KAIST Differential Equations and Applications) Using the method of Green's functions, find the solution of the initial-value problem

$$y'' - 10y' + 25y = e^{5t}, \quad y(0) = -1, \quad y'(0) = 1.$$

Problem 11. (2021 Spring KAIST Differential Equations and Applications) Find a function ϕ such that the function $y = \phi(\ln x)$ satisfies the differential equation

$$x^3y''' + 2x^2y'' - xy' + y = \frac{1}{x}, \quad x > 0.$$

Find the general solution of the equation.

Problem 12. (2021 Spring KAIST Differential Equations and Applications) Find the general solution of the system of second-order differential equations

$$\begin{cases} x'' = -2x + y + t - 1, \\ y'' = 4x - 2y + 3 + \cos(2t). \end{cases}$$

Problem 13. (2021 Spring KAIST Differential Equations and Applications) For a piecewise continuous function $f(t)$ defined on $[0, \infty)$, we have studied in class that a sufficient condition guaranteeing the existence of the Laplace transform $\mathcal{L}\{f(t)\}(s)$ is that f be of exponential order. Meanwhile, there may be a piecewise continuous function that possesses the Laplace transform but is not of exponential order. It is known that the function

$$g(t) = 2te^{t^2} \cos(e^{t^2})$$

is not of exponential order. Nevertheless, show that $\mathcal{L}\{g(t)\}(s)$ exists. (*Hint. Use integration by parts.*)

Problem 14. (2021 Spring KAIST Differential Equations and Applications) For each $t \geq 0$, let

$$f(t) = \int_0^\infty \frac{\cos(t\eta)}{1 + \eta^2} d\eta.$$

Show that

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty \frac{s}{(1 + \eta^2)(s^2 + \eta^2)} d\eta = \frac{\pi}{2(s + 1)}.$$

(*Hint. Consider changing the order of integration.*)

Problem 15. (2021 Spring KAIST Differential Equations and Applications) Find the series solution of the initial-value problem

$$y'' + x^2 y' + y = 0, \quad y(0) = 2, \quad y'(0) = 3,$$

up to the first four nonzero terms.

Problem 16. (2021 Spring KAIST Differential Equations and Applications) Find the general solution of the differential equation

$$4x^2 y'' - 8x^2 y' + (4x^2 + 1)y = 0.$$

Note that $x = 0$ is a singular point of the differential equation. (*Hint. When you obtain a three-term recurrence relation, consider an appropriate substitution that converts the three-term relation into a two-term relation.*)

Problem 17. (2021 Spring KAIST Differential Equations and Applications) Solve

$$\mathbf{x}' = \begin{pmatrix} -3 & -1 & -5 \\ -2 & 1 & -2 \\ 2 & 1 & 4 \end{pmatrix} \mathbf{x}$$

by diagonalization.

Problem 18. (2021 Spring KAIST Differential Equations and Applications) Find the general solution of the nonhomogeneous linear system

$$\mathbf{x}' = \begin{pmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2t + 1 \\ 3 \\ 0 \end{pmatrix}.$$

Problem 19. (2021 Spring KAIST Differential Equations and Applications) Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 4 & 0 \end{pmatrix}.$$

- (a) Find $e^{t\mathbf{A}}$.
- (b) Solve the initial-value problem

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Problem 20. (2021 Spring KAIST Differential Equations and Applications) Consider the linear system

$$\mathbf{x}' = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} \mathbf{x}.$$

- (a) Decide whether the origin is a stable critical point, an asymptotically stable critical point, or an unstable critical point for the system.
- (b) Find the general solution.
- (c) Find the solution $\mathbf{x}(t)$ such that

$$\mathbf{x}(0) = \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}.$$

Problem 21. (2021 Spring KAIST Differential Equations and Applications) Find the solution of the system

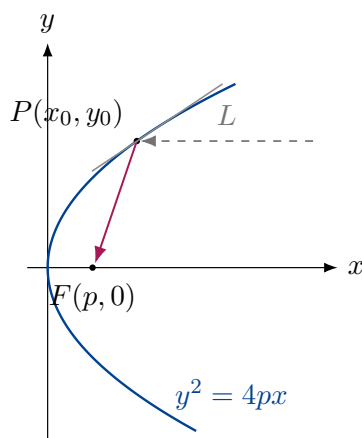
$$\begin{cases} x' = -5x + 4y, \\ y' = -2x + 4y, \end{cases}$$

and draw the phase plane near the critical point $(0, 0)$. Include the direction of some solution curves.

Problem 22. (2021 Spring KAIST Differential Equations and Applications) Find all the critical points of the following nonlinear system of differential equations and classify (if possible) whether they are stable (unstable) nodes or saddle points, spirals, etc.:

$$\begin{cases} x' = 0.2x(1 - 0.2x - 0.3y), \\ y' = 0.5y(1 - 0.1x - 0.2y). \end{cases}$$

Problem 23. (2021 Spring KAIST Differential Equations and Applications) Parabolas have the reflective property that whenever an incident ray parallel to the axis of a parabola is reflected on the parabola, then it will pass through the focus; the following figure depicts a case of the parabola $y^2 = 4px$ ($p > 0$).



If we model such a reflective property as a differential equation, then we obtain the first-order nonlinear differential equation

$$y(y')^2 - 2(p - x)y' - y = 0.$$

More precisely, a solution $y = f(x)$ of the above differential equation is intended to satisfy the following: if an incident ray L parallel to the x -axis is reflected at the point $P(x_0, y_0)$ on the graph of $y = f(x)$, then it passes through the point $F(p, 0)$.

- Verify that the parabola $y^2 = 4px$ satisfies the given differential equation.
- Find a one-parameter family of solutions of the given differential equation. (*Hint. Use the substitution $w = y^2$. Then differentiate the modified equation with respect to the independent variable x .*)
- Characterize a common geometric property of the one-parameter family of solutions you obtained.

Problem 24. (2022 Spring KAIST Differential Equations and Applications) Solve the differential equation

$$y' + y = 1.$$

Problem 25. (2022 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$y' = y^2, \quad y(0) = 1.$$

- Solve the initial-value problem.
- Find the maximal possible interval on which the solution $y(x)$ satisfying $y(0) = 1$ exists.

Problem 26. (2022 Spring KAIST Differential Equations and Applications) Consider a solution $y(x)$ of the autonomous differential equation

$$y' = y(y - 1).$$

- Find all constant solutions.
- Sketch the graph of the solution $y(x)$, $-\infty < x < \infty$, such that $y(0) = \frac{1}{2}$. You do not need to find the formula for $y(x)$.

- (c) Let $y(x)$ be the solution such that $y(0) = \frac{1}{2}$. Find

$$\lim_{x \rightarrow -\infty} y(x) \quad \text{and} \quad \lim_{x \rightarrow +\infty} y(x).$$

You do not need to find the formula for $y(x)$.

- (d) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = 2$. You do not need to find the explicit formula for $y(x)$.
- (e) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = -1$. You do not need to find the explicit formula for $y(x)$.

Problem 27. (2022 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$(x^2 + y^2)y' + (3x^2 + 2x)y = -y^3.$$

- (a) Rewrite the given differential equation as

$$M(x, y) \, dx + N(x, y) \, dy = 0$$

for some functions $M(x, y)$ and $N(x, y)$.

- (b) Show that

$$\frac{\partial M(x, y)}{\partial y} - \frac{\partial N(x, y)}{\partial x} = 3N(x, y).$$

- (c) Let $\mu(x) = e^{3x}$. Show that

$$\frac{\partial(\mu(x)M(x, y))}{\partial y} - \frac{\partial(\mu(x)N(x, y))}{\partial x} = 0.$$

- (d) Consider the equation

$$\mu(x)M(x, y) \, dx + \mu(x)N(x, y) \, dy = 0,$$

and find $f(x, y)$ satisfying

$$\frac{\partial f(x, y)}{\partial x} = \mu(x)M(x, y), \quad \frac{\partial f(x, y)}{\partial y} = \mu(x)N(x, y).$$

- (e) Express the solution of the given differential equation as

$$f(x, y) = C,$$

where C is an arbitrary constant.

Problem 28. (2022 Spring KAIST Differential Equations and Applications) Solve the initial-value problem

$$\frac{dy}{dx} = \cos^2(x - y), \quad y(0) = \frac{\pi}{4}.$$

- (a) Let $u = x - y$. Express du/dx in terms of dy/dx .

- (b) Show that

$$\frac{du}{dx} = \sin^2 u.$$

- (c) Solve the differential equation

$$\frac{du}{dx} = \sin^2 u.$$

- (d) Show that

$$y = x + \operatorname{arccot}(x + C),$$

where C is a constant.

- (e) Determine the constant C such that $y(0) = \pi/4$.

Problem 29. (2022 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$(x^3 + y^3) dx + xy^2 dy = 0.$$

- (a) Let $y = ux$. Rewrite the given differential equation in terms of x and u .
 (b) Find $A(x)$ and $B(u)$ such that

$$A(x) dx + B(u) du = 0.$$

- (c) Solve the differential equation

$$A(x) dx + B(u) du = 0.$$

- (d) Express the solution of the given differential equation in the form

$$g(x, y) = C,$$

where C is a constant.

Problem 30. (2022 Spring KAIST Differential Equations and Applications) Show that the functions

$$y_1(x) = x, \quad y_2(x) = x \ln x, \quad y_3(x) = x(\ln x)^2$$

are linearly independent on $(0, \infty)$, where $\ln$ denotes the natural logarithm.

Problem 31. (2022 Spring KAIST Differential Equations and Applications) Consider the second-order differential equation

$$y'' + y' - 2y = e^x \sin x + 2x - 1.$$

- (a) Find the complementary solutions of

$$y'' + y' - 2y = 0.$$

- (b) Find a particular solution of

$$y'' + y' - 2y = e^x \sin x.$$

- (c) Find a particular solution of

$$y'' + y' - 2y = 2x - 1.$$

- (d) Find a particular solution of

$$y'' + y' - 2y = e^x \sin x + 2x - 1.$$

- (e) Find the general solution of

$$y'' + y' - 2y = e^x \sin x + 2x - 1.$$

Problem 32. (2022 Spring KAIST Differential Equations and Applications) We will use the Green's function to solve the initial-value problem

$$y'' + y' + y = e^{-x/2}, \quad y(0) = 0, \quad y'(0) = 1.$$

- (a) Find the auxiliary equation for the homogeneous equation

$$y'' + y' + y = 0.$$

- (b) Find two independent solutions $y_1(x)$ and $y_2(x)$ of the homogeneous equation.
 (c) Compute the Wronskian of $y_1(x)$ and $y_2(x)$.
 (d) Find the Green's function $G(x, t)$.
 (e) Using the Green's function, find a particular solution $y_p(x)$ of

$$y'' + y' + y = e^{-x/2}$$

that satisfies $y_p(0) = y_p'(0) = 0$.

- (f) Find a solution $y_h(x)$ of

$$y'' + y' + y = 0, \quad y(0) = 0, \quad y'(0) = 1.$$

- (g) Find a solution of

$$y'' + y' + y = e^{-x/2}, \quad y(0) = 0, \quad y'(0) = 1.$$

Problem 33. (2022 Spring KAIST Differential Equations and Applications) Consider the Laplace transform $\mathcal{L}\{f(t)\}(s)$ of functions $f(t)$, $0 \leq t < \infty$.

- (a) Compute the Laplace transform of $f(t) = e^t$ using the definition of the Laplace transform.
 (b) Assume that (i) $f(t)$ is differentiable, (ii) $\mathcal{L}\{f(t)\}(s)$ exists, and (iii) $\mathcal{L}\{f'(t)\}(s)$ exists. Prove that

$$\mathcal{L}\{f'(t)\}(s) = -f(0) + s\mathcal{L}\{f(t)\}(s).$$

- (c) Take $f(t) = e^t$ in part (b), and find the Laplace transform of $f(t) = e^t$. (*Hint. The answer should coincide with the result from part (a).*)
 (d) Solve the differential equation

$$y''(t) + y(t) = 0, \quad y(0) = a_0, \quad y'(0) = a_1,$$

on $(0, \infty)$ by using the Laplace transform.

Problem 34. (2022 Spring KAIST Differential Equations and Applications) For each $t \geq 0$, let

$$f(t) = \int_0^\infty \frac{\sin(tx)}{x} dx$$

and

$$I(x) = \int_0^\infty \sin(tx)e^{-st} dt.$$

- (a) Show that for $s > 0$,

$$\int_0^\infty f(t)e^{-st} dt = \int_0^\infty \frac{1}{x} I(x) dx.$$

- (b) Show that

$$I(x) = \frac{1}{x} - \frac{s^2}{x^2} I(x)$$

for $s > 0$ and $x > 0$.

- (c) Show that

$$I(x) = \frac{x}{x^2 + s^2}.$$

- (d) Show that

$$\mathcal{L}\{f(t)\}(s) = \frac{\pi}{2s}.$$

- (e) Show that

$$f(t) = \frac{\pi}{2}$$

for all $t > 0$.

Problem 35. (2022 Spring KAIST Differential Equations and Applications) Let $y(t)$ be a solution of the differential equation

$$ty'' + y' + ty = 0,$$

or equivalently

$$(ty')' + ty = 0,$$

such that $y(0) = 1$. It is called the Bessel function of the first kind of order $\nu = 0$. Let

$$h(t) = ty'(t), \quad F(s) = \mathcal{L}\{y(t)\}(s), \quad G(s) = \mathcal{L}\{y'(t)\}(s).$$

(a) Show that

$$s\mathcal{L}\{h(t)\}(s) - h(0) - F'(s) = 0.$$

(b) Show that

$$-sG'(s) - F'(s) = 0.$$

(c) Show that

$$G'(s) = F(s) + sF'(s).$$

(d) Show that

$$\frac{F'(s)}{F(s)} = -\frac{s}{s^2 + 1}.$$

(e) Show that

$$F(s) = \frac{C}{\sqrt{s^2 + 1}}.$$

(f) Determine C , and find $\mathcal{L}\{y(t)\}(s)$.

Problem 36. (2022 Spring KAIST Differential Equations and Applications) Let

$$y = \sum_{n=0}^{\infty} a_n x^n$$

be a nonzero series solution of the differential equation

$$y'' - (x^2 - 1)y = 0$$

about the point $x = 0$.

(a) Find the constants p and q such that for $n = 2, 3, 4, \dots$,

$$(pn + q)((n + 2)a_{n+2} + a_n) = na_n + a_{n-2}.$$

(b) Find the series solution satisfying

$$y(0) = 1, \quad y'(0) = 0.$$

Problem 37. (2022 Spring KAIST Differential Equations and Applications) Let

$$\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 1 & 0 \end{pmatrix}.$$

(a) Solve the homogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

(b) Solve the nonhomogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \begin{pmatrix} -\sin t \\ \sin t \end{pmatrix}.$$

Problem 38. (2022 Spring KAIST Differential Equations and Applications) Diagonalize the matrix

$$\mathbf{A} = \begin{pmatrix} -3 & -1 & -5 \\ -2 & 1 & -2 \\ 2 & 1 & 4 \end{pmatrix}.$$

(a) Solve the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

(b) Compute $e^{t\mathbf{A}}$. An answer as a product of three matrices is acceptable.

Problem 39. (2022 Spring KAIST Differential Equations and Applications) Let

$$\mathbf{A} = \begin{pmatrix} 5 & -6 \\ -2 & 1 \end{pmatrix}.$$

(a) Find a fundamental matrix $\Phi(t)$ of the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

(b) Find a particular solution of the nonhomogeneous system

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \begin{pmatrix} 4 \\ 11 \end{pmatrix}.$$

(c) Solve the initial-value problem

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \begin{pmatrix} 4 \\ 11 \end{pmatrix}, \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Problem 40. (2022 Spring KAIST Differential Equations and Applications) Find two linearly independent series solutions, up to the first four nonzero terms, for the differential equation

$$2xy'' + 3y' - xy = 0.$$

You can leave products of integers without multiplying them. For example, the expression

$$\frac{1}{1 \cdot 3 \cdot 5 \cdot 7}$$

is good enough.

Problem 41. (2022 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = x(1 - x - y), \\ y' = y\left(\frac{3}{4} - y - \frac{x}{2}\right). \end{cases}$$

(a) Find the critical points.

(b) Find the Jacobian matrix at each critical point.

(c) Classify the critical points.

Problem 42. (2022 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = 2x - y, \\ y' = 5x - 2y. \end{cases}$$

- (a) Find all the critical points of the system.
- (b) Show that $(0, 0)$ is a center.
- (c) Find

$$\left(\frac{dx}{dt}, \frac{dy}{dt} \right)$$

at the points $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$.

- (d) As t increases, does the point $(x(t), y(t))$ move around $(0, 0)$ in the clockwise or counter-clockwise orientation?

Problem 43. (2022 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = -x + 2y - 4x^3, \\ y' = x - 4y. \end{cases}$$

- (a) Find and classify the critical points of the system.
- (b) Explain why the given system does not have a periodic solution. (*Hint. If there were a periodic solution, then consider a closed curve C and apply Green's theorem. Or, you may use the theorem given in the textbook.*)
- (c) Show that for any constant $a > 0$, the region

$$D = \{(x, y) \mid x^2 + y^2 < a\}$$

is an invariant region. (*Hint. Take a point (x, y) on the boundary of D . Then $(-x, -y)$ has the same direction as the normal vector pointing inside D .*)

- (d) If $(x(t), y(t))$ is a solution satisfying the initial condition

$$(x(0), y(0)) = (x_0, y_0),$$

then what is the long-term behavior of $(x(t), y(t))$ as $t \rightarrow +\infty$?

Problem 44. (2022 Fall KAIST Differential Equations and Applications) For each $x > 0$, let

$$f(x) = \begin{cases} \cos x \sin x, & x \leq \pi, \\ -2 \sin x, & x > \pi. \end{cases}$$

Solve the differential equation

$$y' \cos x + y \sin x = f(x).$$

Find the solution satisfying $y(4\pi/3) = 0$ defined on the largest interval.

Problem 45. (2022 Fall KAIST Differential Equations and Applications) Find the one-parameter family of solutions of the differential equation

$$(y - x)y' + y = 1.$$

Determine whether it is the general solution. Find the solution satisfying $y(1) = 0$ defined on the largest interval.

Problem 46. (2022 Fall KAIST Differential Equations and Applications) Solve the differential equation

$$(1 + 2e^{-y^2}x^2)xyy' = \frac{1}{2}.$$

Find the solution satisfying

$$y(-2) = -\sqrt{\ln 4}$$

defined on the largest interval.

Problem 47. (2022 Fall KAIST Differential Equations and Applications) Find a function of the form $\phi(x) = ax^2 + bx + c$ satisfying the differential equation

$$(x^2 - 1)x^2y'' - (x^2 + 1)xy' + (x^2 + 1)y = 0.$$

Find the general solution of the equation. Find all solutions satisfying the boundary conditions

$$y(-1) = 2, \quad y(1) = -2,$$

if any.

Problem 48. (2022 Fall KAIST Differential Equations and Applications) Find an integer n such that the function $\phi(x) = x^n$ satisfies the differential equation

$$x(x-1)y'' + (3x-2)y' + y = 0.$$

Find the general solution of the equation. Find the solution satisfying

$$y(1/2) = 0, \quad y'(1/2) = 4.$$

Find all solutions satisfying

$$y(1/2) = \ln 4, \quad y(2) = 0,$$

if any.

Problem 49. (2022 Fall KAIST Differential Equations and Applications) Use the Green's function to solve the initial-value problem

$$y'' - 6y' + 9y = e^{4x}, \quad y(0) = 1, \quad y'(0) = -1.$$

Problem 50. (2022 Fall KAIST Differential Equations and Applications) Find the general solution to the system

$$\begin{cases} x' = y, \\ y' = z, \\ z' = w, \\ w' = x, \end{cases}$$

where x, y, z, w are all functions of t .

Problem 51. (2022 Fall KAIST Differential Equations and Applications) Solve the Cauchy–Euler equation

$$x^3y''' + 6x^2y'' + 7xy' + y = 0$$

on $(0, \infty)$.

Problem 52. (2022 Fall KAIST Differential Equations and Applications) Solve

$$y'''(y'')^{-1/2}e^{-x} = 1$$

with initial conditions

$$y(0) = y'(0) = 0, \quad y''(0) = 1.$$

Problem 53. (2022 Fall KAIST Differential Equations and Applications) Solve the initial-value problem

$$y''' - y = 1, \quad y(0) = y'(0) = y''(0) = 0$$

with Laplace transform.

Problem 54. (2022 Fall KAIST Differential Equations and Applications) Let $q > 0$ be a constant. For each $t \geq 0$, let

$$f(t) = \begin{cases} \frac{2t}{q}, & 0 \leq t \leq \frac{q}{2}, \\ 2\left(1 - \frac{t}{q}\right), & \frac{q}{2} \leq t < q, \end{cases}$$

and $f(t+q) = f(t)$. Show that

$$\mathcal{L}\{f(t)\}(s) = \frac{2}{qs^2} \tanh\left(\frac{qs}{4}\right).$$

Problem 55. (2022 Fall KAIST Differential Equations and Applications) Solve the initial-value problem

$$xy'' - 2(1-x)y' - 2(1-x)y = 2e^{-x} \cos x, \quad y(0) = 0, \quad y'(0) = -1.$$

Problem 56. (2022 Fall KAIST Differential Equations and Applications) Find the series solutions of the differential equation

$$(1-x^2)y'' - 2xy' + 6y = 0$$

about the point $x = 0$.

Problem 57. (2022 Fall KAIST Differential Equations and Applications) Let $\nu > 0$ be a constant. Let $\lambda = 1$ or -1 . Find the series solutions of the differential equation

$$x^2y'' - 2xy' + (2 - \lambda\nu^2x^2)y = 0$$

about the point $x = 0$. Identify the elementary functions represented by the series.

Problem 58. (2022 Fall KAIST Differential Equations and Applications) Find the series solutions of the differential equation

$$-x(1+x)^2y'' + (x^2-1)y' + (1-x)y = 0$$

about the point $x = -1$. Identify the elementary functions represented by the series.

Problem 59. (2022 Fall KAIST Differential Equations and Applications) Consider the linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 2 & 1 & 0 \end{pmatrix}.$$

- (a) Find the eigenvalues of $\mathbf{A}$.
- (b) Find a matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is diagonal.
- (c) Let $\mathbf{x} = \mathbf{P}\mathbf{y}$. Construct a system of differential equations for $\mathbf{y}$.
- (d) Solve $\mathbf{y}$.
- (e) Solve $\mathbf{x}$.

Problem 60. (2022 Fall KAIST Differential Equations and Applications) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find $\mathbf{A}^k$, where k is an arbitrary positive integer.
- (b) Find $e^{t\mathbf{A}}$, where t is an arbitrary real number.
- (c) Solve the linear system $\mathbf{x}' = \mathbf{A}\mathbf{x}$.

Problem 61. (2022 Fall KAIST Differential Equations and Applications) A bead with mass m slides along a curve

$$z = x^n e^{-x}.$$

Here n is a positive constant and x takes values on $(0, \infty)$. The damping force is $-\beta v$, where $\beta > 0$ is a constant and v is the velocity of the bead.

- (a) Write down the system of differential equations describing the model.
- (b) Find and classify the critical points of the system.

Problem 62. (2022 Fall KAIST Differential Equations and Applications) Write

$$x'' = \ln \left(\frac{1}{2} + x^2 + 2xx' + (x')^2 \right)$$

as a system of differential equations. Then find and classify the critical points of the system.

Problem 63. (2022 Fall KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = -x^2 + 2y^2, \\ y' = -y^3 + y. \end{cases}$$

- (a) Find the critical point (x_0, y_0) that is in the first quadrant.
- (b) Show that if a constant $c > 0$ is small enough, then

$$\left\{ (x, y) \mid (x - x_0)^2 + (y - y_0)^2 < c^2 \right\}$$

is an invariant region. (*Hint. The change of variables $u = x - x_0$, $v = y - y_0$ may make the computation easier.*)

Problem 64. (2023 Spring KAIST Differential Equations and Applications) Determine whether each of the following statements is true or false.

- (1) Every separable first-order equation

$$\frac{dy}{dx} = g(x)h(y)$$

is exact.

- (2) The substitution $u = y - 2x + 3$ makes the differential equation

$$\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$$

separable.

- (3) A population of bacteria growing exponentially doubles in population every 10 days. Therefore, the population size is described by the differential equation

$$\frac{dy}{dt} = \frac{1}{5}y,$$

where $t = 1$ corresponds to 1 day.

- (4) A model for the population $P(t)$ of the fishery at time t is given by

$$\frac{dP}{dt} = -(P - 400)(P - 1).$$

With this model, the fishery population becomes extinct.

- (5) The initial-value problem

$$xy'' - y' = 0, \quad y(0) = 0, \quad y'(0) = 1$$

has a unique solution.

Problem 65. (2023 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$y' + y = x.$$

- (a) First, solve the differential equation $y' + y = 0$.
 (b) Find a particular solution of the form

$$y_p(x) = Ax + B$$

satisfying $y'_p + y_p = x$, where A and B are constants.

- (c) Find the general solution of $y' + y = x$.
 (d) Find a solution of $y' + y = x$ satisfying $y(0) = 2$.

Problem 66. (2023 Spring KAIST Differential Equations and Applications) Consider a solution $y(x)$ of the differential equation

$$y' = y^2 - 1.$$

- (a) (True/False) If the solution y satisfies $y(0) = \frac{1}{2}$, then y cannot be a constant function.
 (b) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = -\frac{3}{2}$. You do not need to find the formula for $y(x)$.
 (c) (True/False) For the solution y in part (b), $\lim_{x \rightarrow \infty} y(x)$ exists but is not -1 .
 (d) Sketch the graph of the solution $y(x)$ near $x = 0$ such that $y(0) = 0$. You do not need to find the explicit formula for $y(x)$.
 (e) For the solution y in part (d), show by direct computation that

$$\lim_{x \rightarrow \infty} y(x) = -1.$$

Problem 67. (2023 Spring KAIST Differential Equations and Applications) Consider a solution $y(x)$ of the differential equation

$$y' = e^{-y}.$$

- (a) Find the solutions $y(x)$ satisfying $y(0) = -1$, $y(0) = 0$, and $y(0) = 1$.
- (b) Sketch the solutions $y(x)$ satisfying $y(0) = -1$, $y(0) = 0$, and $y(0) = 1$.
- (c) Define $z(x) = e^{-y(x)}$. Show that

$$z' = -z^2.$$

Problem 68. (2023 Spring KAIST Differential Equations and Applications) A tank contains 1000 liters of a solution consisting of 100 kg of salt dissolved in water. Pure water is pumped into the tank at the rate of 5 liters per second, and the mixture, kept uniform by stirring, is pumped out at the same rate. How long will it take until only 10 kg of salt remains in the tank?

Problem 69. (2023 Spring KAIST Differential Equations and Applications)

- (a) (True/False) $\ln x$ and $\ln(2x)$ are both solutions of

$$y'' + \frac{1}{x}y' = 0$$

on $(0, \infty)$.

- (b) (True/False) $\ln x$ and $\ln(2x)$ are linearly independent on $(0, \infty)$.
- (c) For $k \in \{4, 5, 6, \dots\}$, compute the Wronskian of

$$\ln x, \ln(2x), \dots, \ln(kx)$$

on $(0, \infty)$.

- (d) For $k \in \{4, 5, 6, \dots\}$, determine whether

$$\ln x, \ln(2x), \dots, \ln(kx)$$

are linearly independent on $(0, \infty)$. Justify your claim.

- (e) (True/False) $\ln x$, $\ln(2x)$, and $\ln(3x)$ are all solutions of

$$y''' + \frac{2}{x}y'' = 0$$

on $(0, 1)$.

- (f) (True/False) $\ln x$, $\ln(2x)$, and $\ln(3x)$ are linearly independent on $(0, \infty)$.
- (g) (True/False) If $k \in \{4, 5, 6, \dots\}$, then $\ln x, \ln(2x), \dots, \ln(kx)$ are all solutions of

$$y^{(k)} + \frac{k-1}{x}y^{(k-1)} = 0$$

on $(0, \infty)$.

Problem 70. (2023 Spring KAIST Differential Equations and Applications) Find the general solution of the differential equation

$$y^{(4)} - 4y^{(3)} + 8y'' - 8y' + 4y = e^{-x}(5x + 4\sin x).$$

Problem 71. (2023 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$x^2 y'' - 2xy' + x^2 y = x^4. \quad (1)$$

(a) Let

$$T(y) = x^2 y'' - 2xy' + x^2 y.$$

Calculate $T(\cos x)$, $T(\sin x)$, $T(x \cos x)$, and $T(x \sin x)$.

(b) Find a fundamental set of solutions of the associated homogeneous equation

$$T(y) = 0.$$

You may use the property

$$T(ay_1 + by_2 + cy_3 + dy_4) = aT(y_1) + bT(y_2) + cT(y_3) + dT(y_4)$$

for real numbers a, b, c, d and functions y_1, y_2, y_3, y_4 .

(c) Find the general solution of (1).

Problem 72. (2023 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$x^2 y'' + 5xy' + 5y = 0, \quad x > 0.$$

(a) (*True/False*) There is a real number r such that x^r is a solution of the equation.

(b) Let a be a positive constant. Find the solution of the equation subject to

$$y(a) = -a^2, \quad y'(a) = 2a.$$

Problem 73. (2023 Spring KAIST Differential Equations and Applications) Consider

$$y'' + 2y' + y = x$$

with conditions

$$y(0) = 0 \quad (2)$$

and

$$y(1) = 0. \quad (3)$$

(a) (*True/False*) There is a nonconstant solution of the corresponding homogeneous equation satisfying conditions (2) and (3).

(b) Find the general solution of the corresponding homogeneous equation.

(c) Find two linearly independent solutions of the corresponding homogeneous equation such that one of them satisfies (2) and the other satisfies (3).

(d) Find the Green's function $G(x, t)$ of the two solutions you found.

(e) (*True/False*)

$$\phi(x) = \int_0^1 G(x, t) t \, dt$$

is a particular solution of the above differential equation but does not satisfy $\phi(0) = \phi(1)$.

(f) Find a solution of the original nonhomogeneous equation subject to conditions (2) and (3).

Problem 74. (2023 Spring KAIST Differential Equations and Applications) Set

$$f(t) = \begin{cases} \sin t, & t < \frac{\pi}{2}, \\ \cos t, & t \geq \frac{\pi}{2}. \end{cases}$$

- (a) Write $f(t)$ in terms of the Heaviside function, i.e., the unit step function.
 (b) Find the Laplace transform of $f(t)$.
 (c) Solve

$$y''(t) - y(t) = f(t)$$

subject to $y(0) = 0$ and $y'(0) = 1$.

Problem 75. (2023 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = y + t^2, \\ y' = z + t, \\ z' = x, \end{cases}$$

where x , y , and z are all functions of t .

- (a) Use the method of elimination to obtain a differential equation involving only x and t .
 (b) Find $x(t)$.
 (c) Use the expression for $x(t)$ and the first equation in the system to find $y(t)$.
 (d) Use the expression for $y(t)$ and the second equation in the system to find $z(t)$.

Problem 76. (2023 Spring KAIST Differential Equations and Applications) Determine whether each of the following statements is true or false.

- (1) Since $x = 0$ is an irregular singular point of

$$x^3 y'' - xy' + y = 0,$$

the differential equation possesses no solution that is analytic at $x = 0$.

- (2) The general solution of

$$x^2 y'' + xy' + (x^2 - 1)y = 0$$

is

$$y = c_1 J_1(x) + c_2 J_{-1}(x),$$

where $J_1(x)$ and $J_{-1}(x)$ are Bessel functions of the first kind of orders 1 and -1 , respectively.

- (3) The Bessel function of the second kind $Y_\nu(x)$ is linearly independent of $J_{-\nu}(x)$ for noninteger ν .
 (4) The superposition principle for solutions of a system of linear equations holds only for homogeneous cases.
 (5) The general solution of the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

with a real symmetric $n \times n$ matrix $\mathbf{A}$ is always written using the eigenvalues and eigenvectors of $\mathbf{A}$.

Problem 77. (2023 Spring KAIST Differential Equations and Applications) Let a and b be two nonzero real constants. For $t \geq 0$, let

$$f_a(t) = e^{at} \quad \text{and} \quad f_b(t) = e^{bt}.$$

- (a) Assume that $a = b$. Compute $(f_a * f_a)(t)$ for $t \geq 0$.
 (b) Assume that $a \neq b$. Compute $(f_a * f_b)(t)$ for $t \geq 0$.
 (c) (True/False) For all a, b , there exist $M > 0$ and $c > 0$ such that

$$|(f_a * f_b)(t)| \leq M e^{ct}$$

for all $t \geq 0$.

- (d) Prove that

$$\lim_{b \rightarrow a} (f_a * f_b)(t) = (f_a * f_a)(t)$$

for all $t \geq 0$.

- (e) (True/False) For $b \neq a$,

$$(\cos bt * \cos at)(t) = \frac{\sin bt - \sin at}{b^2 - a^2}$$

for all $t \geq 0$.

Problem 78. (2023 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$y''(t) + 2y'(t) + 5y(t) = 0$$

with $y(0) = 2$ and $y'(0) = 3$. In the following, solve the differential equation with the given initial conditions using the Laplace transform.

- (a) Apply the Laplace transform $\mathcal{L}$ to the above differential equation and obtain

$$p(s)\mathcal{L}\{y(t)\}(s) - q(s) = 0,$$

where $p(s)$ and $q(s)$ are polynomials in s . Find $p(s)$ and $q(s)$.

- (b) Find the function $h(t)$ whose Laplace transform is

$$\frac{q(s)}{p(s)}.$$

- (c) Check that $h(t)$ is a solution of the above differential equation.
 (d) (True/False) For a Riemann-integrable function $f(t)$,

$$y(t) = \frac{1}{2} \int_0^t e^{-\tau} \sin(2\tau) f(t - \tau) \, d\tau$$

is a solution of

$$y''(t) + 2y'(t) + 5y(t) = f(t), \quad y(0) = y'(0) = 0, \quad t \geq 0.$$

Problem 79. (2023 Spring KAIST Differential Equations and Applications) Consider the differential equation

$$xy'' + 2y' + xy = 0, \quad x > 0. \tag{1}$$

- (a) Classify the singular points, if any.

- (b) Let

$$T(y) = xy'' + 2y' + xy.$$

Compute $T(x^{n+r})$ for a natural number n and a real number r .

- (c) Find the conditions on
- c_n
- and
- r
- such that

$$\sum_{n=0}^{\infty} c_n x^{n+r}$$

is a solution of equation (1), i.e.,

$$T\left(\sum_{n=0}^{\infty} c_n x^{n+r}\right) = \sum_{n=0}^{\infty} c_n T(x^{n+r}) = 0.$$

- (d) Find the series solutions of equation (1) and show that they form a fundamental set of solutions of equation (1).
 (e) Find the smallest $l > 0$ such that

$$-\infty < \lim_{x \rightarrow 0^+} x^l y(x) < \infty$$

for every y that satisfies equation (1), and prove it.

Problem 80. (2023 Spring KAIST Differential Equations and Applications) Find the solution of the system

$$\mathbf{x}' = \begin{pmatrix} 0 & -1 \\ 13 & 4 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 0 \\ 3 \end{pmatrix}. \quad (2)$$

Let

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 13 & 4 \end{pmatrix}.$$

- (a) (True/False) $\mathbf{A}$ has a real eigenvalue.
 (b) Find the matrix exponential $e^{t\mathbf{A}}$ by using the Laplace transform and show that

$$e^{(t+s)\mathbf{A}} = e^{t\mathbf{A}} e^{s\mathbf{A}}$$

for all $t, s > 0$.

- (c) Find the solution of system (2).
 (d) (True/False) Let $\mathbf{x}(t)$ be the solution of (2). Suppose $\mathbf{y}(t)$ is the solution of

$$\mathbf{y}' = \begin{pmatrix} 0 & -1 \\ 13 & 4 \end{pmatrix} \mathbf{y}, \quad \mathbf{y}(0) = e^{\pi} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Then there exists $t > 0$ such that

$$\mathbf{y}(t) \neq \mathbf{x}\left(t + \frac{\pi}{2}\right).$$

Problem 81. (2023 Spring KAIST Differential Equations and Applications) Consider the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \begin{pmatrix} 1 & -3 & 3 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix}.$$

- (a) Find the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\mathbf{A}$.
 (b) Find a set of linearly independent vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ satisfying

$$\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i, \quad \|\mathbf{v}_i\| \neq 0, \quad i = 1, 2, 3. \quad (3)$$

- (c) (True/False) Suppose $\{\mathbf{v}_i\}_{i=1}^3$ satisfies (3). Let

$$\mathbf{x} = \sum_{i=1}^3 x_i \mathbf{v}_i.$$

Then $\mathbf{x}' = \mathbf{A}\mathbf{x}$ is equivalent to

$$x'_i(t) = \lambda_i x_i(t), \quad i = 1, 2, 3.$$

- (d) Find the solution to the system satisfying the initial condition

$$\mathbf{x}(0) = \begin{pmatrix} 1 \\ -3 \\ -1 \end{pmatrix}.$$

Problem 82. (2023 Spring KAIST Differential Equations and Applications) Consider the linear system

$$\mathbf{x}' = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{-2t} \sec^3 t.$$

Let

$$\mathbf{A} = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix}.$$

- (a) Compute $e^{t\mathbf{A}}$.
 (b) (True/False) $e^{t\mathbf{A}}$ is invertible for all $t \geq 0$.
 (c) (True/False)

$$\mathbf{u}(t) = \int_0^t e^{-s\mathbf{A}} \begin{pmatrix} 2 \\ 0 \end{pmatrix} (e^{-2s} \sec^3 s) \, ds$$

is a solution of the above linear system for $t \geq 0$.

- (d) Find the general solution of the above linear system.

Problem 83. (2023 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = x - 2y, \\ y' = 2x - y. \end{cases}$$

- (a) Classify the critical point of this system.
 (b) Find the velocity (x', y') at the point $(c, 0)$, where $(c, 0)$ is an arbitrary point on the positive half of the x -axis of the xy -plane.
 (c) Draw the curve of a nontrivial solution $(x(t), y(t))$ on the xy -plane, with an arrow on it to show the direction of motion of the point $(x(t), y(t))$ as t increases.

Problem 84. (2023 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = -2x + 2x^2, \\ y' = -3x + y + 3x^2. \end{cases}$$

- (a) Find the critical points of this system.
- (b) Find the Jacobian matrix at each critical point.
- (c) For each critical point, find the trace and determinant of the Jacobian matrix.
- (d) Classify each critical point.

Problem 85. (2023 Spring KAIST Differential Equations and Applications) Consider the system

$$\begin{cases} x' = y, \\ y' = -x + y(1 - x^2 - y^2). \end{cases} \quad (4)$$

- (a) Find the critical points of this system.
- (b) Suppose $(x(t), y(t))$ is a solution of (4). Let

$$r(t) = x(t)^2 + y(t)^2.$$

Derive an equation satisfied by

$$r' = 2xx' + 2yy'.$$

- (c) Suppose $r(0) > 1$. Prove that

$$r(t) < r(0)$$

for all $t > 0$ by using the equation derived in part (b).

- (d) (True/False) The region

$$R = \{(x, y) \mid x^2 + y^2 < c\}$$

is an invariant region of Type I when $c > 1$.

- (e) Suppose $c > 1$ is a constant. Show that there is at least one periodic solution in the region

$$\{(x, y) \mid x^2 + y^2 < c\}.$$